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ALTITUDE AND ENVIRONMENTAL RESPONSES OF ENTRIES IN THE 1972-73 EASTERN AFRICAN MAIZE VARIETY TRIAL[1]

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Regional maize (*Zea mays* L.) variety trials provide a means of assessing gains in yield potential and adaptation of breeding populations over a range of altitudes and environments. Such trials allow existing commercial hybrids and varieties to be compared with experimental ones. Results from past trials [1, 6] have showed that Kenya's hybrids 512 and 632 and Zambia's hybrid SR52 were well adapted to environments that ranged from 0.9 to 1.6 km above sea level. Kenya's hybrids 611C and 613B were superior at altitudes above 1.6 km. The mean response to altitude in the 1968-69 Eastern African Maize Variety Trial (EAMVT) [6] was 17 q/ha/km, whereas in the 1970-71 EAMVT it was 13 q/ha/km [1]. Adaptation was related to altitudinal and environmental responses that were greater for varieties and hybrids selected at or originating from high altitudes.

Lower mean temperatures are the primary effect of high altitude, though in Eastern Africa high altitudes are also associated with one long rainy season [7]. Diseases, such as *Helminthosporium turcicum* (Pass.) (blight) and *Puccinia sorghi* (Schw.) (rust) often limit yields of lowland material grown at environments over 1.2 km altitude.

Materials and Methods

The 1972-73 EAMVT consisted of 26 supplied entries and two local varieties. Seven countries entered varieties, and the EAC-EAAFRO Maize Genetics programme in Kenya entered four. Unfortunately, germination of Bambu from Rwanda was extremely poor, and it was dropped before continuing with the analyses presented in this paper. Katumani Composite B, a short-season commercial entry from Kenya, was subject to excessive theft that resulted in poor data. It was also omitted from these analyses.

Table I lists the entries, their sources, and pedigrees if available. Most of the composites have been formed and are undergoing recurrent selection as described by Eberhart, Harrison, and Ogada [5]. Included were entries to evaluate progress from selection within Ecuador 573 and Kitale II [2, 3]. Two substrains of both Ukirigu Composite B and Kawanda Composite A were evaluated.

The trial was grown as three replications of a randomized complete block design. A five-row plot was used, with data being taken on the three centre rows, except for the date of black layer formation which was obtained from the border rows because the sampling was destructive. Plots were overplanted and thinned at 30 cm to the desired stand, leaving two plants in the end hills and one in between. A spacing of 75 cm between rows and 30 cm between plants was commonly used. Rates and methods of fertilizer application were consistent with recommendations from collaborating stations.

Data were recorded on shelled grain weights and the kernel moisture per cent, plant stand, lodging, bare tips, usable ears, diseased ears, date of tasseling, date of black layer formation, ear heights, and blight score. Yields were adjusted to 12.5 per cent moisture and by covariance, to the mean stand per cent for the site. Stand per cent was calculated as (counted stand/planted stand) × 100. Lodging, diseased ear, and usable ear counts were expressed as a per cent of the counted stand. The number of ears with bare tips was expressed as a per cent of the usable ears. Days

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to tassel were taken counted from the date of planting to 50 per cent pollen shed. Days of grain filling were from the date of tasseling to the date of black layer formation. Ear heights were recorded as the distance from the soil surface to the top ear leaf collar for an average of 10 competitive plants. Blight susceptibility was scored between 0.0 for no symptoms and 5.0 for very seriously affected. Rust data were not recorded.

Co-operating sites were divided into three groups according to altitude; low, from 0.0-0.85 km; medium, from 0.90-1.60 km; and high, above 1.65 km. Table II lists the sites and their altitude, mean yield, and coefficient of variation.

The stability analysis of Eberhart and Russell [4] was applied by first fitting altitude as the independent variable (Model 1) and then altitude and mean site yield (Model 2) as the independent variables. Sites were considered as random environments for this analysis.

$$\text{Model 1: } Y_{ij} = m_i + b'A_iA_j + e_{ij}$$

$$\text{Model 2: } Y_{ij} = m_i + bA_iA_j + bI_iI_j + e_{ij}$$

In the models, Y_{ij} is the observed yield of the ith variety in the jth environment (site); m_i is the mean yield of the ith variety; bA_i is the response of the ith variety per km of altitude; A_j is the deviation of the altitude of the jth environment from the mean altitude of all environments; bI_i is the response of the ith variety per unit change of the environmental index; I_j, the environmental index, is the difference between the mean of the jth environment and the mean for all environments; and e_{ij} is the deviation from linear fit.

Results and Discussion

The correlation of altitude and mean site yield was 0.53 and highly significant for all sites combined, but this correlation was not significant for any individual altitude group. Estimates of b_A and b_I were statistically independent within groups, because a wide range of environments were sampled in each. Mean yields ranged from 9.6 q/ha at Kibaha, Tanzania (0.10 km), to 99.4 q/ha at Mulungwishi, Zaire (1.20 km). Coefficients of variation were high in some cases, but this was generally due to very low yields.

Analysis of variance for yield are shown in Table III for each altitude group and for all sites combined. The effect of altitude was not significant within any of the altitude groupings, but overall it had a significant effect. Differences among varieties and the sites x varieties (environment x genotype) interaction were significant in all analyses. Partitioning of the site x variety interaction showed the varieties x altitude[1] and varieties x environments[1] (adj. alt.) interactions to be significant in the medium, high, and combined analyses but not in the low altitude analysis. All residuals and pooled deviations were highly significant. When variation due to varieties x altitute[1] and varieties x environments[1] (adj. alt.) is removed from the sites x varieties interaction, the estimate of σ^2_{GE} is reduced from 75.3 to 40.8. [Note: σ^2_{GE} = (sites x varieties mean square —pooled error mean square) or σ^2_{GE} = (pooled deviations mean square — pooled error mean square) after partitioning. No division by the number of replications was done since the analysis was performed on means.] Similar reductions have been found in previous results [1, 6] and indicate that nearly half of the sites x varieties interaction can be explained in terms of differing responses of varieties to altitude and environments. There was no such reduction in the low altitude analysis and only slight reductions in the medium and high altitude analyses.

Results from fitting Models 1 and 2 for the yield data are given in Table IV. The combined response to altitude alone was 19.3 q/ha per km. Varieties that were significantly more responsive than average in the combined analysis were H5022, H613(R)C3, H611(R)C3, H611C, H613C, H611D, and H611, all of which were above 29.5 q/ha per km. Among the least responsive were MAS, IC, and H5011. The same pattern occurred in the medium and high group analyses. In the low altitude analysis, Kw.CA(M)Cl and MAS were the most responsive. The high altitude varieties had a relatively lower response pattern at low altitude sites.

Fitting Model 2 gave estimates of the response to environment. It should be noted that if $b_{I_i} = 1.0$, $b_{A_i} = b'_{A_i} - \bar{b}'_A$ where $\bar{b}'_A$ is the mean response to altitude alone. In the combined analysis, H613(R)C3, H622, SR52, H611(R)C3, and UCA × UCB−F had estimates of b_I exceeding 1.1 q/ha per unit increase in the index. H5011, H511, and IC had

responses significantly lower than 1.0. The deviation mean squares were greatly reduced by fitting b_1, indicating that Model 2 would be preferable for predicting performance of the varieties. UCA × UCB−W would be the most stable over all environments, followed by H622 and Kw.CA(S)C1. H5011, H611D, and H613(R)C3 were the least stable, where stability is taken as minimum deviation from a linear environmental response.

Due to the wide range of environments and altitudes in this trial, evaluation of varietal performance should be based on the appropriate altitude group analysis. At low altitudes, UCA(F)C5, SR52, H5033, and H632 responded at 1.2 or above for the environmental index. UCA(F)C5, Kw.Ca(M)C1, CCA, and KCB × KCE had small deviation mean squares. The best yielding variety, MAS, was significantly better than all others except IC. Mean yields of the varieties that responded well to the environmental index were significantly lower than MAS or IC, except for H5033 which was not different from IC.

The environmental responses of H613(R)C3, H622, and H611(R)C3 were significantly greater than 1.0 at the medium altitude. Of these, H622 and H611(R)C3 had reasonably low deviations from linearity. The best mean yield was obtained from SR52 (55.6 q/ha) although 10 other varieties did not differ significantly.

No variety's environmental response was significantly higher than 1.0 at the high altitudes, although H5022 at 1.35 was meaningful. Generally, the deviation mean squares were higher than in the low and medium altitude analyses, partly due to higher mean yield levels. Five varieties with high yields were not significantly different: H5022, H613(R)C3, H611(R)C3, H613C, and H611D. H5022 significantly outyielded the sixth ranking variety, H611C.

Considering the combined data from 59 sites, three cycles of reciprocal recurrent selection improved the yield of the variety-cross hybrid H611 by 27 per cent (H611 vs. H611(R)C3), or 9 per cent per cycle. The environmental response of H611(R)C3 was significantly greater than that of original H611. H613(R)C3, using the third cycle of Ec573(R12), was significantly higher yielding than H613C, which used the second cycle. No yield difference existed between H611C and H611D. The yield performance of UCA(F)C5 was indistinguishable from its crosses to UCB. Both the mass and S_1 selection versions of Kw.CA yielded identically and were significantly lower than the cross of Kw.CA to MCI. Embu (1 × II) yielded no differently than H511, which it is intended to replace, and both were significantly outyielded by H512.

Days to tassel and days of grain filling were analysed to determine the effect of altitude. Table V contains the analyses of variance, with the mean squares for altitude[1] and varieties x altitude[1] being significant for both characters. Although the residual mean squares were significant, much of the site variation was accounted for by differences in altitude.

The responses to altitude for days to tassel and days of grain filling are given in Table 6. The mean delay in tasseling was 22.6 days per km of altitude. The range of response was from 20.8 days for H511 to 24.9 days for CCA. The mean number of days to tasseling were shortest for H511 (73) and lowest for H611C (86). Many of the low and medium altitude varieties, such as ZCA, H632, H622, MCI x Kw.CA, CCA, MAS, IC, and H5033, showed a significantly greater delay in tasseling than did high altitude varieties like the H611 crosses and H613C.

Errors in determining the days of grain filling were considerably larger than those for days to tasseling (Table V). Data from several sites had nonsignificant F ratios even with large apparent differences among means. The mean length of the grain filling period increased by 20.4 days per km increase in altitude with the greatest response shown by H611 at 24.8 days (Table VI). Least responsive were IC, MAS, H632, H5011, H5033, and CCA. The number of days of grain filling over all sites was greatest for H613C, H5022, H611D, H611, and H611(R)C3, though these could not be statistically differentiated from many other varieties. The correlations between yield and days of grain filling, and between yield and the response to altitude of days of grain filling were not significant in the combined data from all sites. However, considering only the high altitude sites, where varietal differences in days

of grain filling were significant, a correlation of 0.7** was obtained between yield and days of grain filling.

Generally, there was a good agreement between varieties, their maturity, and the sites where they would normally be grown. A single long rainy season requires a maize that will mature at the end of the rain, not earlier. In Eastern Africa, where field drying is the practice, a variety which matures before the end of the rains is subject to ear rots and severe stalk lodging. Long season varieties yield poorly in environments having short rainy seasons and, there, varieties such as MAS, IC, and H5011 are preferable.

Agronomic data for the low, medium, and high altitude sites are shown in Tables VII, VIII and IX. At sites from low altitudes (Table VII), MAS had the lowest incidence of root lodging and fewest diseased ears. H5011 had significantly more usable ears than all other entries except MAS. It required 61 days for the average variety to tassel. H511 was the earliest at 55 days and the H611 series latest at 68 days. Differences in grain filling were not detected, and there was an average of 54 days from tasseling to black layer formation.

SR52 had significantly less root lodging than many varieties at the medium altitude sites (Table VIII). No varietal differences were significant for stalk lodging. H5011, MAS, IC, CCA, Kw.CA(M)Cl, and SR52 had fewer than 8 per cent bare tips. H5011 had significantly more usable ears (113.3 per cent) than any other variety. MAS and IC, at 95.5 and 94.3 ears per 100 plants, were significantly more prolific than all remaining varieties except MCI × Kw.CA. Days to tassel ranged from 64 for H511 to 77 for H611C with a mean of 72. The ear height of H5011 at 122 cm was significantly lower than that of any other variety. Ear heights of H611C, H611D, and H611 exceeded 203 cm and were significantly different from the fourth highest variety, H611(R)C3. ZCA, CCA, MAS, and IC were more susceptible to blight than the medium and high altitude hybrids developed in Kenya. No significant differences among varieties were found for days of grain filling.

The agronomic data from high altitude sites (Table IX) showed H5011 to have good root and stalk lodging resistance. H611(R)C3 had the lowest incidence of stalk lodging at 12.2 per cent. Stalk lodging in H622 and H632 was significantly greater than that in H611(R)C3. H5033, MAS, and IC had worse stalk lodging than most varieties, which agreed with their being too short maturing for use in highland areas. Bare tips were significantly more common in H511 than other varieties. Usable ears were greatest for H5011 at 105.6 per cent, with many of the higher-yielding varieties not significantly different from H5011. The number of diseased ears was relatively high on H5033, ZCA, and H622. The average number of days to tassel was 97, with H611C, H611D, CCA, H611, and H613C requiring more than 100 days. H511, H512, and Embu (I × II) showed fewer days to tassel than most varieties. H611(R)C3 at 198 cm ear height was significantly lower than H611C, H611, and H611D. Varieties with ear heights below 150 cm were SR52, H511, and H5011. Blight affected ZCA, H5011, MAS, and IC more than other varieties. Differences among varieties in the number of days of grain filling were significant and ranged from 78 for IC to 94 for H613C, H611, H611(R)C3, H611D, and H5022.

Predicted yields at the mean elevation and mean yield for low, medium, and high altitude sites are shown in Table X. Model 2, using b_A and b_I from the combined analysis, was evaluated to obtain the predictions. For example, the predicted yield of SR52 at 1.18 km and a mean yield level of 50.1 q/ha was calculated as follows:—

$$Y = 48.8 - 3.6\ (1.18 - 1.16) + 1.15\ (50.1 - 46.5) = 52.9 \text{ q/ha},$$

where 48.8 was the mean yield of SR52, 3.6 was bA, 1.16 was the mean altitude of all sites, 1.15 was bI, and 46.5 was the mean yield of all sites. When the mean yield level of the high altitude sites was raised to 80 q/ha., predicted yields for H613(R)C3, H5022, and H611(R)C3 exceeded 95 q/ha.

The correlations of observed and predicted yields were very high and significant, indicating the utility and accuracy of the prediction equation. When selecting varieties for commercial use, data from these many sites are more reliable than data from a limited number of sites in one or two years.

Summary

The 1972-73 Eastern African Maize Variety Trial of 26 entries was successfully grown at

59 sites at elevations of from 10 m to 2,250 m. Stability analyses were done using altitudes and mean site yields as independent variables. The reduction in size of the deviation mean squares indicated that fitting altitude alone was not as good as fitting altitude plus environmental index. The mean response to altitude was 19.3 q/ha per km altitude, with H5022, H613(R)C3, H611(R)C3, H611C, H611D, H611, and H613C responding above 29 q/ha per km in the overall analysis. H613(R)C3, H622, SR52, and H611(R)C3 had the highest environmental index response when both altitude and index were fitted.

At sites from 0.0-0.85 km, MAS and IC were the best yielders. In the medium altitude zone of 0.9-1.6 km, there were no significant differences among SR52, H622, H5033, H613(R)C3, MCI × Kw.CA, UCA × UCB-F, H5022, UCA × UCB-W, H512, H632, or H611(R)C3. The best yielding varieties above 1.6 km were H5022, H613(R)C3, H611(R)C3, H613C and H611D.

The response of days to tasseling to altitude was a 22.6 day delay per km altitude. A similar delay of 20.4 days was calculated for the regression of altitude on days of grain filling. A positive significant correlation of 0.7** was found for days of grain filling and yield at the high altitude sites. Correlations of observed and predicted yields using data from 59 sites were highly significant. Use of the regression coefficients to predict and select appropriate varieties for given environments was encouraged.

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TABLE I—ENTRIES IN THE 1972–73 EASTERN AFRICAN MAIZE VARIETY TRIAL

Entry	Entry abbreviation	Source	Pedigree or type of material
SR52	SR52	Zambia (commercial)	N3233 x SC5522
Zambia Composite A	ZCA	Zambia	Composite
H511	H511	Kenya (commercial)	Embu Comp. 11 x Embu Comp. 12
H512	H512	Kenya (commercial)	H511 x SR52
H632	H632	Kenya (commercial)	(A x F) G
H622	H622	Kenya (commercial)	(A x F) (G x D)
H613C	H613C	Kenya (commercial)	(F x G) Ecuador 573 (R12)C2
H611C	H611C	Kenya (commercial)	Kitale III x Ecuador 573(R12)C2
H611	H611	EAC *	Kitale II x Ecuador 573
H611(R)C3	H611 (R) C3	EAC *	Kitale II (R11)C3 x Ecuador 573 (R12)C3
Kitale III x Ecuador 573(R12)C3	H611D	EAC *	Variety cross
H613(R)C3	H613 (R) C3	EAC *	(F x G) Ecuador 573 (R12) C3.
H5022	H5022	Kenya (Kenya Seed Co.)	Closed pedigree
Kitale Comp. B(S)C2 x Kitale Comp. E(S)C2	KCB x KCE	Kenya	Composite cross
Ukiriguru Comp. A(F)C5 x Ukiriguru Comp. B—Full	UCA x UCB-F	Tanzania	Composite cross
Ukiriguru Comp. A(F)C5 x Ukiriguru Comp. B—White (F)C3	UCA x UCB-W	Tanzania	Composite cross
Ukiriguru Composite A(F)C5	UCA(F)C5	Tanzania	Composite
H5011	H5011	Kenya (Kenya Seed Co.)	Closed pedigree
Kawanda Composite A(S)C1	Kw.CA(S)C1	Uganda	Composite
Kawanda Composite A(M)C1	Kw.CA(M)C1	Uganda	Composite
Mexican Comp. I x Kawanda Comp. A	MCI x Kw.CA	Uganda	Composite cross
Chitedze Composite A	CCA	Malawi	Composite
Medium Altitude selection	MAS	Tanzania	(SV28 x IC) Selected
Ilonga Composite	IC	Tanzania	Composite
H5033	H5033	Kenya (Kenya Seed Co.)	Closed pedigree
Embu Comp. I x Embu Comp. II	Embu (I x II)	Kenya	Composite cross
Local entry	Local	—	—
Local entry	Local	—	—

*EAC—EAAFRO Maize Genetics Division, Kenya.

TABLE II—ALTITUDE (KM), MEAN YIELDS (Q/HA), AND COEFFICIENT VARIATION (C.V.%) OF SITES IN THE 1972–73 EASTERN AFRICAN MAIZE VARIETY TRIAL

Low (0–0·85 km)				Medium (0·90–1·60 km)				High (1·65 km up)			
Site	Mean			Site	Mean			Site	Mean		
	Alt.	Yield	C.V.%		Alt.	Yield	C.V.%		Alt.	Yield	C.V.%
Afgoi, Somalia ..	0·01	33·3	39	Suluti, Tanzania ..	0·90	46·8	20	Rubona (72), Rwanda	1·65	26·1	33
Kikambala, Kenya ..	0·02	11·1	52	Mparambo, Burundi	0·90	16·4	24	Rubona (73), Rwanda	1·65	37·7	25
Chambezi, Tanzania ..	0·05	29·5	28	Aduku, Uganda ..	0·91	33·3	28	Bako, Ethiopia ..	1·65	83·2	13
Makanga, Malawi ..	0·05	20·8	29	Makokha, Malawi ..	1·00	66·5	17*	Deneba, Ethiopia ..	1·68	77·2	20
Kibaha, Tanzania ..	0·10	9·6	32*	Mitalula, Tanzania ..	1·05	33·9	30	Areka, Ethiopia ..	1·70	29·4	21*
Mtwara, Tanzania ..	0·12	15·5	38	Namulonge, Uganda	1·07	21·2	28	Awassa, Ethiopia ..	1·70	68·4	14
Mokwa, Nigeria ..	0·15	24·1	29*	Abbi, Uganda ..	1·07	42·1	20	Melko, Ethiopia ..	1·75	45·8	24
Yousafwala, Pakistan..	0·21	44·7	10	Serere, Uganda ..	1·07	71·8	21	Mbeya, Tanzania ..	1·77	75·6	10
Kilombero, Tanzania..	0·34	34·0	24	Bukalasa, Uganda ..	1·12	34·4	16	Njombe, Tanzania ..	1·80	62·8	14
Samaru, Nigeria ..	0·37	42·0	27	Chitedze, Malawi ..	1·15	75·5	15	Kisii, Kenya.. ..	1·82	53·1	18
Wami, Tanzania ..	0·43	20·0	24	Kisanga, Zaire ..	1·19	91·5	13	Saboti, Kenya ..	1·85	47·3	31
Nachingwea, Tanzania	0·43	26·2	26	Ukiriguru, Tanzania	1·20	12·3	42	Endebess, Kenya ..	1·86	75·8	15
Baka, Malawi	0·50	34·8	20	Mulungwishi, Zaire..	1·20	99·4	15	Endebess, Kenya ..	1·86	58·0	19
Ilonga, Tanzania ..	0·51	24·4	41	Likasi, Zaire.. ..	1·20	59·5	20	Kitale, Kenya ..	1·87	56·7	21
Chitala, Malawi ..	0·60	30·3	26	Nakabango, Uganda	1·21	38·3	35	Kitale, Kenya ..	1·87	55·9	32
				Mt. Makulu, Zambia	1·21	28·3	26	Alemaya, Ethiopia ..	2·08	72·0	16
Mean	0·26	26·7		Mt. Makulu, Zambia	1·21	24·7	32	Kulumsa, Ethiopia ..	2·17	66·3	15*
r = 0·35				Copper Belt, Zambia	1·25	43·9	17	Igeri, Tanzania ..	2·24	22·0	21
				Mosso, Burundi ..	1·25	40·7	22	Mt. Elgon, Kenya ..	2·25	75·5	15
				Didesa, Ethiopia ..	1·30	93·9	13				
				Ismani, Tanzania ..	1·35	65·3	7	Mean	1·85	57·3	
				Bele, Ethiopia ..	1·35	79·2	31	r = 0·05			
				Abela, Ethiopia ..	1·40	57·3	23				
				Embu, Kenya ..	1·46	24·4	52				
				Kakamega, Kenya ..	1·58	52·8	15				
				Mean	1·18	50·1					
				r = 0·21							

*Approximate.
r = 0·53** over all sites.

TABLE III—ANALYSES OF VARIANCE FOR YIELD (Q/HA) OF 26 ENTRIES IN THE 1972–73 EASTERN AFRICAN MAIZE VARIETY TRIAL

Source	Low altitude sites		Medium altitude sites		High altitude sites		Combined	
	df	Mean square	df	Mean square	df	Mean square	df	Mean square
Sites	14	2,773·8**	24	16,220·3**	18	8,889·3**	58	13,924·9**
Altitude[1]	1	4,668·9	1	17,890·1	1	486·5	1	242,325·2**
Residual	13	2,628·1**	23	16,147·7**	17	9,383·6**	57	10,233·7**
Varieties	25	503·2**	25	409·0**	25	1,988·6**	25	818·2**
Sites × varieties	350	44·8**	600	67·4*	450	113·8**	1,450	109·9**
Varieties × altitude[1]	25	40·9	25	133·3**	25	178·8*	25	1,848·2**
Residual	325	45·1**	575	64·5**	425	110·0**	1,425	79·4**
Varieties × env.[1] (adj. Alt.)	25	34·6	25	265·5**	25	168·2*	25	303·0**
Pooled deviations	300	45·9**	550	55·4**	400	106·3**	1,400	75·4**
Pooled error†	810	19·6	1,269	41·4	1,026	38·5	3,186	34·6

*Indicates significance at 5%.

**Indicates significance at 1%.

[1]Linear.

†Based on full trial of 28 entr ies; harmonic mean used to obtain the sum of squares on a mean basis; df are approximate due to missing plots

TABLE IV—REGRESSION COEFFICIENTS AND DEVIATION MEAN SQUARES (D.M.S.) OBTAINED BY FITTING MODEL 1 AND MODEL 2 FOR YIELD (Q/HA) DATA IN THE 1972–73 EASTERN AFRICAN MAIZE VARIETY TRIAL. b'A AND bA ARE RESPONSES IN Q/HA PER KM ALTITUDE, AND bI IS THE RESPONSE PER UNIT INCREASE IN THE INDEX

Variety	Low (0–0·85 km), 15 sites						Medium (0·90–1·60 km), 25 sites					
	Mean Yield	Model 1		Model 2			Mean Yield	Model 1		Model 2		
		b'A	D.M.S.	bA	bI	D.M.S.		b'A	D.M.S.	bA	bI	D.M.S.
SR52	26·9	22·8	172·4	0·8	1·25	16·6	55·6	32·9	831·5	−2·0	1·11	68·8
ZCA	23·5	20·5	132·8	3·1	0·98	37·8	46·9	42·1	672·9	10·8	0·99	62·0
H511	25·5	12·8	84·2	−1·5	0·81	19·8	47·1	22·5	388·8	−1·2	0·75	37·5
H512	28·7	21·2	94·3	6·0	0·86	20·8	53·0	36·3	678·2	5·2	0·99	74·5
H632	28·8	9·2	164·4	−12·2	1·21	17·5	52·5	34·3	845·3	−1·6	1·14	39·5
H622	30·0	11·2	172·0	−9·7	1·18	33·6	54·9	34·9	922·4	−2·7	1·19	40·1
H613C	21·7	7·9	137·8	−10·5	1·04	30·6	48·4	53·5	795·0	18·9	1·10	46·6
H611C	17·4	2·7	90·4	−11·5	0·80	27·3	46·3	43·7	748·8	10·4	1·06	57·9
H611	14·9	12·8	95·4	0·2	0·71	47·6	38·2	38·7	522·2	12·0	0·86	81·4
H611(R)C3	22·8	23·3	142·1	5·1	1·03	37·5	52·2	41·7	915·9	4·5	1·18	51·9
H611D..	22·3	18·9	136·9	3·2	0·89	61·6	47·8	34·0	823·0	−1·3	1·12	41·3
H613(R)C3	23·7	11·6	131·8	−6·2	1·01	31·3	54·3	45·6	1,077·0	5·8	1·26	88·7
H5022	21·9	20·1	116·2	6·4	0·78	60·0	53·6	43·9	710·8	11·5	1·03	53·8
KCB × KCE.. ..	20·5	12·7	108·7	−4·5	0·97	14·2	43·3	41·7	601·7	11·8	0·95	42·7
UCA × UCB–F ..	28·7	16·6	147·0	−3·3	1·12	21·0	53·9	43·7	729·3	10·2	1·06	27·9
UCA × UCB–W ..	28·0	21·3	89·6	6·9	0·81	24·5	53·3	19·5	760·1	−14·6	1·08	31·8
UCA (F) C5	27·9	21·9	178·7	−1·1	1·30	8·6	50·2	43·5	639·0	12·4	0·99	35·4
H5011	33·1	17·5	326·3	−1·3	1·07	229·2	48·4	−0·2	461·8	−24·5	0·77	96·8
Kw.CA(S)C1 ..	26·5	18·0	134·3	−1·1	1·08	18·0	47·2	33·8	475·0	6·8	0·86	21·0
Kw.CA(M)C1 ..	25·2	36·5	128·4	17·3	1·08	10·1	48·9	36·4	634·6	6·0	0·97	55·9
MCI × Kw.CA ..	30·6	10·4	113·3	−4·8	0·86	42·1	54·0	22·9	559·7	−4·6	0·87	91·3
CCA	23·7	16·9	135·5	−2·7	1·11	12·0	47·8	21·4	566·1	−7·3	0·91	49·4
MAS	41·2	34·6	196·2	18·3	0·92	119·8	50·3	1·8	603·0	−27·8	0·94	59·9
IC	37·2	29·0	239·4	9·9	1·08	131·3	50·9	9·2	443·7	−16·0	0·80	46·6
H5033	33·8	26·3	173·9	4·9	1·21	28·3	54·5	16·5	812·0	−18·7	1·12	38·4
Embu (I × II) ..	29·2	2·9	112·2	−11·7	0·82	47·1	50·0	24·2	543·4	−4·1	0·90	43·8
Mean	26·7	17·7	144·4	0·0	1·00	44·2	50·1	31·5	683·1	0·0	1·00	53·3
L.S.D. 0·05	4·8†	24·5		26·4	0·52		4·1†	26·7		25·3	0·17	

†Based on pooled deviations from Model 2 fit.

TABLE IV—REGRESSION COEFFICIENTS AND DEVIATION MEAN SQUARES (D.M.S.) OBTAINED BY FITTING MODEL 1 AND MODEL 2 FOR YIELD (Q/HA) DATA IN THE 1972–73 EASTERN AFRICAN MAIZE VARIETY TRIAL. b'A AND bA ARE RESPONSES IN Q/HA PER KM ALTITUDE, AND bI IS THE RESPONSE PER UNIT INCREASE IN THE INDEX

Variety	High (1·65 km up), 19 sites						Combined, 59 sites					
	Mean Yield	Model 1		Model 2			Mean Yield	Model 1		Model 2		
		b'A	D.M.S.	bA	bI	D.M.S.		b'A	D.M.S.	bA	bI	D.M.S.
SR52	57·0	−17·0	595·9	−22·9	1·13	146·5	48·8	18·6	602·7	−3·6	1·15	84·0
ZCA	48·4	−14·6	336·2	−19·0	0·84	85·7	41·4	15·8	441·9	−3·0	0·98	68·4
H511	55·1	13·1	346·1	8·4	0·91	52·7	44·2	18·4	285·2	3·1	0·80	36·8
H512	60·8	1·2	436·4	−4·0	0·99	85·0	49·4	20·2	440·9	1·2	0·99	59·0
H632	54·7	−0·0	408·3	−5·0	0·95	88·1	47·2	16·3	527·0	−4·9	1·10	49·7
H622	59·0	8·0	427·2	24·9	1·06	26·1	49·9	18·4	560·2	−3·9	1·16	33·2
H613C	69·9	28·7	604·3	22·7	1·15	135·5	48·5	30·3	544·5	9·5	1·08	86·1
H611C	67·5	11·2	590·8	5·6	1·05	202·9	45·8	30·5	513·7	10·8	1·02	105·0
H611	62·7	27·6	588·2	21·3	1·21	61·7	40·2	29·7	416·7	12·3	0·90	99·1
H611(R)C3	72·2	24·0	580·7	18·2	1·10	151·9	51·2	30·9	577·8	9·3	1·12	82·6
H611D..	69·7	29·7	625·8	24·2	1·05	241·0	48·3	29·5	522·4	9·3	1·05	121·0
H613(R)C3	73·4	33·5	673·0	27·3	1·17	193·1	52·7	31·2	672·5	8·2	1·19	113·1
H5022	75·3	24·9	782·5	17·8	1·35	135·6	52·5	33·2	550·8	12·4	1·08	92·8
KCB × KCE.. ..	58·7	13·1	344·9	8·4	0·91	52·2	42·5	23·9	377·0	6·1	0·92	42·1
UCA × UCB–F ..	57·9	5·6	563·7	−0·6	1·18	64·0	48·8	18·9	522·8	−2·5	1·11	37·6
UCA × UCB–W ..	59·6	0·7	384·1	−4·6	0·99	29·7	48·9	19·5	458·3	−0·6	1·04	30·9
UCA(F)C5	55·9	−2·8	565·8	−9·1	1·19	53·7	46·4	18·0	489·6	−2·7	1·07	38·5
H5011	39·9	17·1	442·0	13·3	0·71	277·3	41·8	5·1	429·1	−9·9	0·78	192·5
Kw.CA(S)C1 ..	53·3	−5·3	361·6	−10·1	0·91	64·6	43·9	16·8	346·3	−0·4	0·89	33·4
Kw.CA(M)C1 ..	52·0	−2·7	397·0	−7·9	1·00	35·7	43·9	17·5	433·7	−1·7	1·00	41·9
MCI × Kw.CA ..	54·9	1·4	410·2	−3·4	0·91	114·8	48·4	15·1	398·9	−2·2	0·90	82·7
CCA	41·7	−23·2	319·3	−27·4	0·81	85·7	39·7	11·0	417·4	−7·1	0·94	72·9
MAS	37·7	−18·0	240·9	−21·4	0·65	94·4	43·9	−1·4	408·2	−18·3	0·88	108·2
IC	43·0	−14·9	262·2	−18·9	0·76	57·1	44·9	4·1	349·8	−11·9	0·83	81·3
H5033	60·2	−1·3	440·9	−6·7	1·04	51·4	51·1	16·4	510·9	−4·7	1·09	40·4
Embu (I × II) ..	50·0	−3·5	404·6	−8·6	0·97	71·2	44·7	12·8	390·7	−5·0	0·93	52·3
Mean	57·3	5·2	466·6	0·0	1·00	102·2	46·5	19·3	468·8	0·0	1·00	72·5
L.S.D. 0·05	6·6†	35·3		34·7	0·36		3·1†	5·1		5·9	0·16	

† Based on pooled deviations from Model 2 fit.

TABLE V—ANALYSES OF VARIANCE FOR DAYS TO TASSEL AND DAYS OF GRAIN FILLING IN THE 1972–1973 EASTERN AFRICAN MAIZE VARIETY TRIAL

Source	Days to tassel		Days of grain filling	
	df	Mean square	df	Mean square
Sites	45	8,637·2**	15	6,826·1**
Altitude	1	262,975·2**	1	63,363·1**
Residual[1]	44	2,856·8**	14	2,787·7**
Varieties	25	559·6**	25	116·1**
Sites × varieties	1,125	17·6**	375	33·6**
Varieties × altitude[1]	25	26·1**	25	58·6*
Pooled deviations	1,100	17·4**	350	31·8**
Pooled error†	2,349	4·5	756	14·5

* Indicates significance at 5%.

** Indicates significance at 1%.

[1] Linear.

† Based on full trial of 28 entries; harmonic mean used to obtain sums of squares on a mean basis; excludes sites not furnishing replicate data; df are approximate due to missing plots.

TABLE VI—MEANS, REGRESSION COEFFICIENTS, AND DEVIATION MEAN SQUARES (D.M.S.) OBTAINED FROM FITTING DAYS TO TASSEL AND DAYS OF GRAIN FILLING TO ALTITUDE IN THE 1972–73 EASTERN AFRICAN MAIZE VARIETY TRIAL. bA IS THE RESPONSE IN DAYS PER KM ALTITUDE

Variety	Days to tassel, 46 sites			Days of grain filling, 16 sites		
	Mean	bA	D.M.S.	Mean	bA	D.M.S.
SR52	76·8	23·4	118·9	77·3	18·9	121·4
ZCA..	78·2	23·2	105·3	76·9	17·4	108·6
H511	72·8	20·8	99·2	78·6	23·3	136·1
H512	75·4	21·9	107·4	78·4	20·7	128·4
H632	82·3	23·7	125·3	73·9	16·1	135·4
H622	80·8	23·7	97·1	76·4	18·9	136·3
H613C	84·6	21·7	112·2	80·5	24·6	131·8
H611C	86·0	21·2	138·5	78·3	24·1	105·5
H611	85·5	21·2	142·0	79·8	24·8	128·9
H611(R)C3	84·0	21·7	113·7	79·6	24·3	119·9
H611D	85·6	23·0	126·2	79·9	22·9	119·7
H613(R)C3	83·8	21·8	105·3	77·1	20·8	108·6
H5022	82·0	24·8	121·2	80·0	23·1	152·5
KCB × KCE	82·1	22·4	114·9	78·6	23·9	126·8
UCA × UCB-F	81·3	24·6	126·0	77·0	19·8	122·2
UCA × UCB-W	80·4	22·8	109·9	78·0	20·1	147·9
UCA(F)C5	82·3	22·9	118·4	76·8	23·6	238·0
H5011	76·9	22·2	121·8	73·9	16·2	257·7
Kw.CA(S)CI	76·8	22·1	135·6	78·3	20·0	144·4
Kw.CA(M)CI	80·9	22·9	118·5	77·6	21·5	139·1
MCI × Kw.CA	80·1	23·5	104·5	72·9	18·3	117·3
CCA	83·7	24·9	130·7	72·0	16·3	112·9
MAS	79·7	23·3	99·9	72·0	15·9	135·6
IC	78·3	23·3	136·7	71·1	15·7	111·8
H5033	81·2	24·0	114·4	74·8	16·2	149·3
Embu (I × II)	75·4	21·0	120·0	77·3	23·2	146·1
Mean	80·6	22·6	117·8	76·8	20·4	137·8
L.S.D. 0·05	1·7†	0·9		3·9†	6·5	

†Based on pooled deviations.

TABLE VII—AGRONOMIC DATA OBTAINED FROM LOW ALTITUDE SITES IN THE 1972–73 EASTERN AFRICAN MAIZE VARIETY TRIAL
(Number of sites is shown in parentheses)

Variety	Yield (q/ha) (15)	Root lodg. % (10)	Stalk lodg. % (7)	Usable ears % (15)	Diseased ears % (14)	Days to tassel (11)	Ear height (cm) (13)	Blight (0·0–5·0) (6)	Days of grain filling (2)
SR52	26·9	42·8	9·9	70·2	13·5	56·7	115·7	1·5	57
ZCA	23·5	44·0	17·6	60·4	12·3	58·5	128·9	1·8	55
H511	25·5	46·7	17·4	72·4	13·0	55·1	126·0	1·5	51
H512	28·7	51·8	12·6	73·8	11·1	56·7	129·9	1·5	53
H632	28·8	44·9	14·1	76·1	13·8	61·8	131·5	1·3	55
H622	30·0	42·5	12·0	72·5	10·2	60·1	132·7	1·0	55
H613C	21·7	46·8	12·6	66·2	11·2	65·6	157·6	0·8	53
H611C	17·4	48·2	10·9	53·3	11·1	68·0	174·5	1·0	53
H611	14·9	46·2	12·4	47·7	9·6	68·0	171·9	0·8	54
H611(R)C3	22·8	45·2	14·8	66·4	10·9	65·3	164·3	1·3	51
H611D	22·3	47·1	12·0	62·7	11·7	65·6	167·8	1·0	57
H613(R)C3	23·7	47·0	11·4	69·8	8·2	65·0	160·8	0·8	55
H5022	21·9	52·5	9·6	62·6	10·6	64·4	150·7	1·3	55
KCB × KCE	20·5	48·8	12·6	58·9	11·2	63·3	151·2	1·3	52
UCA × UCB–F	28·7	45·6	12·9	72·8	11·2	60·0	141·5	1·0	55
UCA × UCB–W	28·0	39·9	11·8	74·5	11·1	60·5	143·1	1·3	54
UCA(F)C5	27·9	39·2	13·7	70·2	11·7	62·5	150·5	1·0	51
H5011	33·1	46·8	12·2	99·4	8·4	57·5	104·8	0·8	55
Kw.CA(S)C1	26·5	42·9	10·9	68·9	10·3	57·7	128·2	1·3	54
Kw.CA(M)C1	25·2	44·7	14·8	65·4	9·3	61·4	144·3	1·3	56
MCI × Kw.CA	30·6	47·3	11·1	76·7	9·2	59·7	140·3	1·5	50
CCA	23·7	42·2	11·6	66·2	9·7	62·2	146·7	1·3	52
MAS	41·2	33·5	11·1	90·6	6·2	59·5	142·3	1·3	53
IC	37·2	36·6	12·2	88·0	9·2	58·5	137·5	0·8	50
H5033	33·8	42·9	11·1	81·6	10·8	60·5	133·0	1·0	51
Embu (I × II)	29·2	42·8	14·4	74·9	12·3	57·1	135·2	1·3	53
Mean	26·7	44·5	12·6	70·8	10·7	61·2	142·7	1·2	53·5
L.S.D. 0·05	4·8†	8·4§	6·5ns§	9·4§	4·1ns§	2·7§	10·9§	0·5§	9·7ns

†Based on pooled deviations from Model 2 fit.

§Approximate.

TABLE VIII—AGRONOMIC DATA OBTAINED AT MEDIUM ALTITUDE SITES IN THE 1972–73 EASTERN AFRICAN MAIZE VARIETY TRIAL
(Number of sites is shown in parentheses)

Variety	Yield (q/ha) (25)	Root lodg. % (21)	Stalk lodg. % (14)	Bare tips % (8)	Usable ears % (20)	Diseased ears % (22)	Days to tassel (17)	Ear height (cm) (20)	Blight (0–5) (14)	Days of grain filling (6)
SR52	55·6	20·1	14·4	7·4	84·2	11·5	67·3	142·8	1·5	69
ZCA	46·9	25·7	16·7	11·0	74·5	16·9	68·8	158·7	1·9	72
H511	47·1	29·8	18·5	18·1	78·8	16·0	63·8	153·9	1·6	71
H512	53·0	30·6	17·4	17·5	81·6	15·1	66·0	149·9	1·5	73
H632	52·5	28·6	20·1	15·1	83·1	15·1	73·6	162·9	1·1	71
H622	54·9	29·8	18·7	11·8	83·3	15·0	71·8	165·3	1·2	69
H613C	48·4	25·5	18·4	17·9	79·2	12·2	75·8	192·7	1·1	71
H611C	46·3	27·0	15·0	15·3	73·0	14·3	77·0	209·7	1·1	68
H611	38·2	28·9	16·9	20·4	64·3	15·4	76·0	203·8	1·3	68
H611(R)C3 ..	52·2	25·8	15·7	11·4	80·7	12·4	75·1	194·2	1·3	71
H611D	47·8	23·9	18·2	13·8	75·2	14·0	76·9	203·8	1·2	70
H613(R)C3 ..	54·3	28·3	18·7	14·0	82·1	13·8	75·2	191·5	1·2	69
H5022	53·6	25·7	15·7	15·9	81·1	13·9	73·3	182·6	1·2	71
KCB × KCE ..	43·3	28·6	18·1	24·8	70·5	17·8	72·7	189·2	1·1	70
UCA × UCB–F ..	53·9	25·7	17·3	11·9	83·7	11·9	72·9	177·0	1·6	69
UCA × UCB–W ..	53·3	26·5	17·2	12·4	81·7	12·2	71·6	173·8	1·7	71
UCA(F)C5 ..	50·2	22·2	14·0	11·1	81·6	13·7	73·8	182·4	1·7	66
H5011	48·4	22·9	14·6	2·5	113·3	9·9	67·8	122·4	1·9	68
Kw.CA(S)C1 ..	47·2	25·1	15·2	12·0	79·3	14·2	67·9	156·9	1·5	72
Kw.CA(M)C1 ..	48·9	27·7	18·2	7·1	78·2	11·4	72·0	179·8	1·6	71
MCI × Kw.CA ..	54·0	30·5	18·8	9·6	88·5	10·5	71·5	172·7	1·8	68
CCA	47·8	27·8	14·8	7·5	84·3	12·4	74·4	178·6	1·9	66
MAS	50·3	25·9	18·4	5·6	95·5	9·1	71·0	171·8	2·1	68
IC	50·9	26·0	18·9	7·5	94·3	9·8	69·1	169·5	1·9	68
H5033	54·5	29·1	18·5	15·4	82·1	12·0	72·1	167·6	1·5	70
Embu (I × II) ..	50·0	29·2	20·7	14·5	83·7	12·0	66·3	161·5	2·0	70
Mean	50·1	26·8	17·3	12·8	82·2	13·2	71·7	173·7	1·5	69
L.S.D. 0·05 ..	4·1†	6·7§¶	5·5¶ns	6·7§	7·6§¶	4·2§¶	2·5§¶	9·4§¶	0·5¶	6¶ns

†Based on pooled deviations from Model 2 fit.

§Approximate.

¶Based on pooled location × entry sums of squares.

TABLE IX—AGRONOMIC DATA OBTAINED AT HIGH ALTITUDE SITES IN THE 1972–73 EASTERN AFRICAN MAIZE VARIETY TRIAL
(Number of sites is is shown in parentheses)

Variety	Yield (q/ha) (19)	Root lodg. % (15)	Stalk lodg. % (15)	Bare tips % (7)	Usable ears % (18)	Diseased ears % (19)	Days to tassel (18)	Ear height (cm) (18)	Blight (0–5) (12)	Days of grain filling (18)
SR52	57·0	11·4	15·3	6·4	84·9	16·9	94·4	148·5	1·4	89
ZCA	48·4	15·2	25·0	6·4	73·7	20·0	95·4	159·7	2·1	87
H511	55·1	15·7	19·9	23·6	90·5	15·7	88·8	147·7	1·3	92
H512	60·8	11·7	17·4	13·4	87·4	11·9	91·9	153·6	1·4	89
H632	54·7	13·8	20·3	9·0	89·2	19·5	99·0	163·7	1·1	82
H622	59·0	11·1	22·4	7·9	98·5	20·6	98·1	165·4	1·1	87
H613C	69·9	15·3	14·7	14·0	95·5	12·6	100·3	196·8	0·8	94
H611C	67·5	14·1	17·0	11·2	91·3	8·6	102·0	210·2	0·9	92
H611	62·7	17·5	14·2	11·0	87·1	10·4	101·1	290·3	1·1	94
H611(R)C3 ..	72·2	12·4	12·2	10·2	96·5	9·9	99·6	198·0	1·3	94
H611D	69·7	11·8	14·8	11·6	95·3	10·7	101·8	207·0	1·1	94
H613(R)C3 ..	73·4	12·2	17·0	11·8	98·7	12·9	99·4	189·5	0·9	89
H5022	75·3	10·0	13·8	9·5	97·2	12·0	96·9	184·4	0·9	94
KCB × KCE ..	58·7	18·2	22·7	12·8	86·6	14·1	98·6	184·8	1·1	92
UCA × UCB–F ..	57·9	14·1	21·4	6·8	88·1	15·6	98·5	176·8	1·3	89
UCA × UCB–W..	59·6	11·4	20·3	9·3	93·3	16·3	96·7	172·1	1·6	89
UCA(F)C5 ..	55·9	11·9	16·6	7·9	86·3	15·3	98·4	179·0	1·3	92
H5011	39·9	6·5	14·1	1·0	105·6	13·6	93·6	106·9	2·1	85
Kw.CA(S)C1 ..	53·3	12·9	20·5	11·5	82·4	15·8	93·1	157·0	1·4	89
Kw.CA(M)C1 ..	52·0	14·3	20·5	6·7	86·4	12·0	97·2	179·0	1·4	89
MCI × Kw.CA ..	54·9	16·8	18·1	8·0	92·0	15·2	96·8	165·0	1·9	82
CCA	41·7	16·6	24·6	4·0	83·5	11·8	101·5	176·4	1·9	82
MAS	37·7	12·9	28·0	5·0	88·8	14·3	96·5	167·3	2·1	80
IC	43·0	10·8	27·5	5·2	91·1	15·7	95·2	159·9	2·2	78
H5033	60·2	11·9	31·5	11·3	99·7	24·8	98·6	169·6	1·1	85
Embu (I × II) ..	50·0	13·9	17·6	11·5	88·3	13·8	91·5	157·0	1·9	89
Mean	57·3	13·2	19·5	9·5	90·7	14·6	97·1	184·0	1·4	88
L.S.D. 0·05 ..	6·6†	5·2	7·3	6·2	13·5§	6·0§	3·1§	8·4	0·5	7§

†Based on pooled deviations from Model 2 fit.

§Approximate.

TABLE X—PREDICTED YIELDS ($\hat{Y}$, Q/HA) AT THE MEAN ELEVATION AND MEAN YIELD FOR THE LOW, MEDIUM, AND HIGH ALTITUDE SITES IN THE 1972–73 EASTERN AFRICAN MAIZE VARIETY TRIAL. VALUES ARE BASED ON bA AND bl FROM THE COMBINED REGRESSION ANALYSES

Variety	Low (0·26 km)		Medium (1·18 km)		High (1·85 km)		
	Y	$\hat{Y}$	Y	$\hat{Y}$	Y	$\hat{Y}$	$\hat{Y}'$
SR52	26·9	29·3	55·6	52·9	57·0	58·7	84·8
ZCA	23·5	24·7	46·9	44·9	48·4	49·9	72·2
H511	25·5	25·6	47·1	47·1	55·1	55·0	73·1
H512	28·7	28·7	53·0	53·0	60·8	60·9	83·4
H632	28·8	29·8	52·5	51·1	54·7	55·7	80·7
H622	30·0	30·4	54·9	54·0	59·0	59·7	86·1
H613C	21·7	18·6	48·4	52·6	69·9	66·7	91·2
H611C	17·4	15·9	46·3	49·7	67·5	64·3	87·4
H611	14·9	11·3	38·2	43·7	62·7	58·4	78·8
H611(R)C3	22·8	20·7	52·2	55·4	72·2	69·7	95·1
H611D	22·3	19·1	47·8	52·3	69·7	66·1	89·9
H613(R)C3	23·7	21·8	54·3	57·1	73·4	71·2	98·2
H5022	21·9	20·0	53·6	56·6	75·3	72·7	97·2
KCB × KCE	20·5	18·8	43·3	45·9	58·7	56·6	77·5
UCA × UCB-F	28·7	29·1	53·9	52·7	57·9	59·1	84·3
UCA × UCB-W	28·0	28·8	53·3	52·6	59·6	59·7	83·3
UCA(F)C5	27·9	27·6	50·2	50·2	55·9	56·1	80·4
H5011	33·1	35·3	48·4	44·4	39·9	43·4	61·1
Kw.CA(S)C1	26·5	26·6	47·2	47·1	53·3	53·2	73·4
Kw.CA(M)C1	25·2	25·6	48·9	47·5	52·0	53·5	76·2
MCI × Kw.CA	30·6	32·6	54·0	51·6	54·9	56·6	77·0
CCA	23·7	27·5	47·8	42·9	41·7	45·0	66·3
MAS	41·2	42·9	50·3	46·7	37·7	40·8	60·8
IC	37·2	39·2	50·9	47·7	43·0	45·7	64·5
H5033	33·8	33·7	54·5	54·9	60·2	59·6	84·4
Embu (I × II)	29·2	30·8	50·0	47·9	50·0	51·3	72·4
Mean	26·7	26·7	50·1	50·1	57·3	57·3	80·0
L.S.D. 0·05†	4·8		4·1		6·6		
r	0·98**		0·75**		0·99**		

†Based on pooled deviations from Model 2 fit.

$\hat{Y}'$ Mean yield set at 80 q/ha.